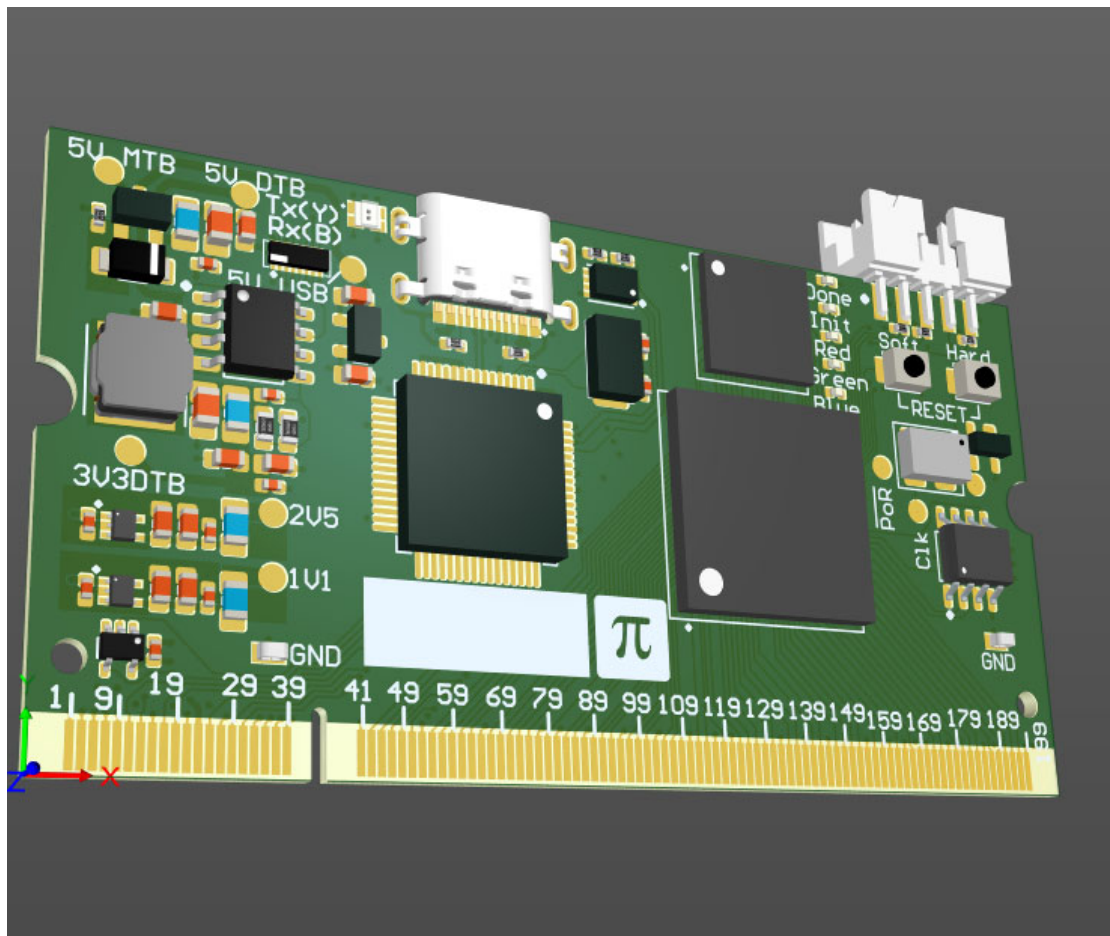


Altium Designer 2024



1. Introduction

1.1 Document Content

This document contains some information about how to start drawing a new Schematic and Pcb using Altium Designer.

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3. New Project

3.1 Software installation

Go on <https://altium.hevsi.ch> and download the install Altium 24 package. It's very important to follow the installation instructions.

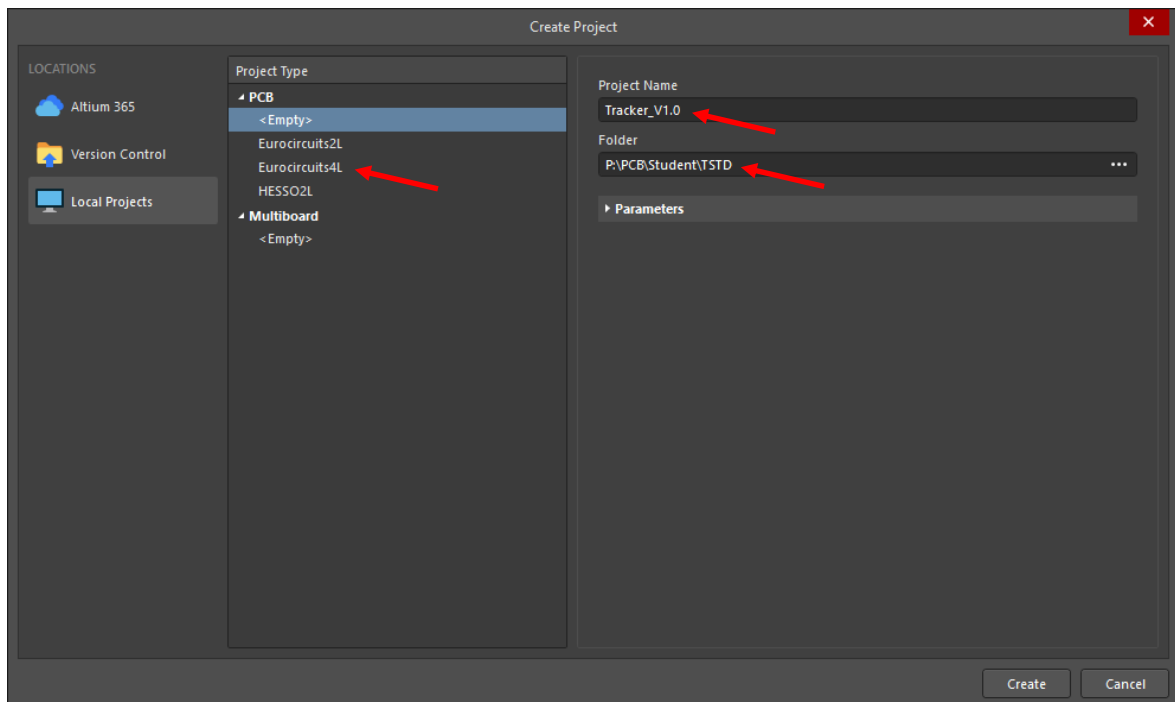
3.2 Creating a new Project

Select *File->New->Project...* and choose one of the templates **HESO2L.PrjPcb**, **Eurocircuit2L.PrjPcb** or **Eurocircuit4L.PrjPcb**.

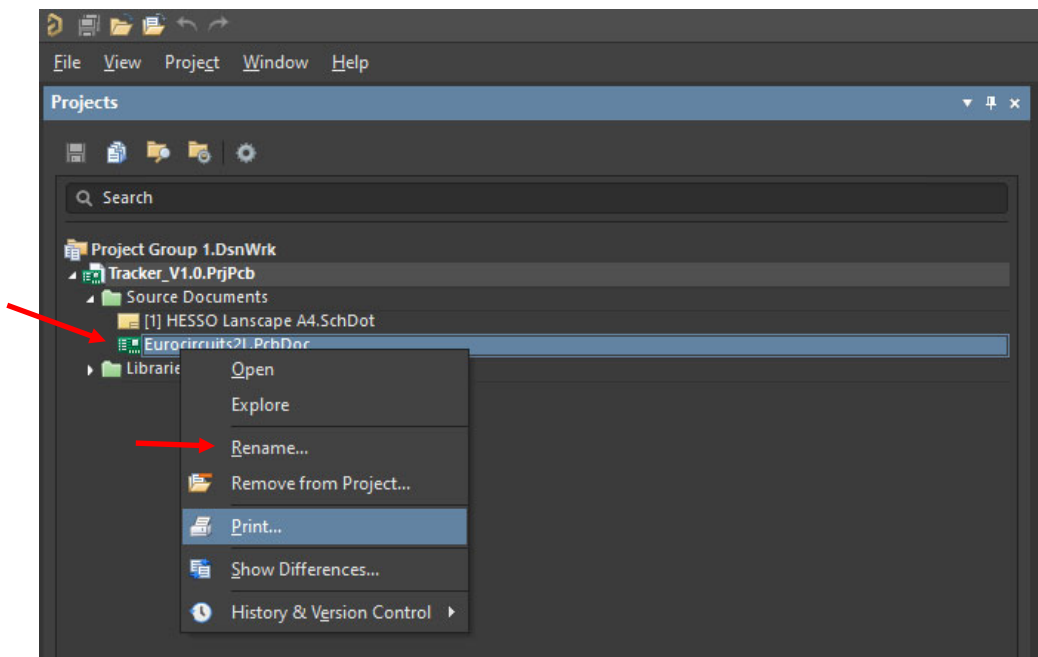
If you're not sure, ask someone in electronic Laboratory 23N219.

Please give an explicit name to your project, **Pcb_project** or **Board_project** aren't. It is very useful to give a version number in the name of your project for ex.: **Tracker_V1.0**

Then choose where you want to save it.



3.3 Rename the project files



Right click on the **.PcbDoc** and rename it. If your project name is **Tracker_V1.0** then give the same name to **.PcbDoc**. Repeat the process for the **.Schdoc** file.

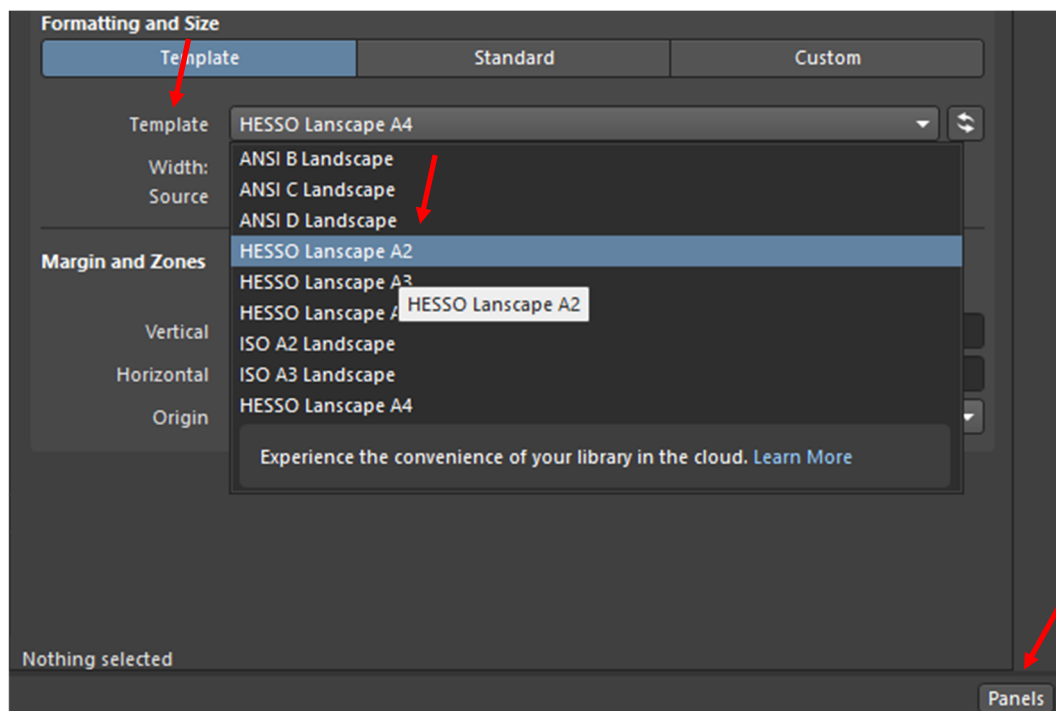
3.4 Change the size of the schematic file

The default schematic is Landscape A4 but there are more choices.

Open the schematic file.

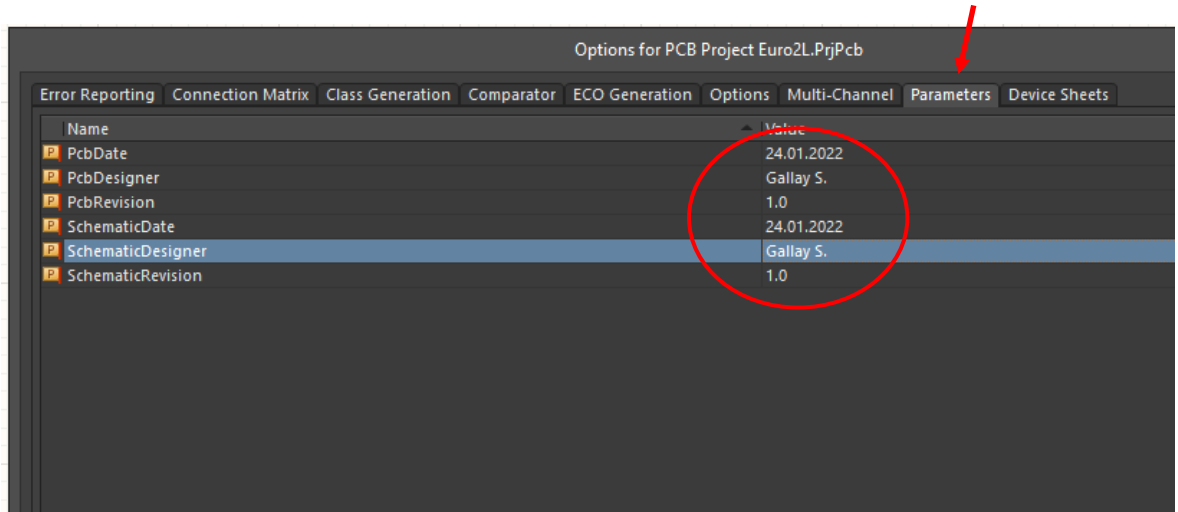
On the right bottom corner, click on the **Panels** button then choose **Properties**.

In the **Template** menu, Select the one you want.



3.5 Defining the Project Parameters

Right Click on *****.PrjPcb** -> *Project Option* -> *Parameter*, and fill all the fields.



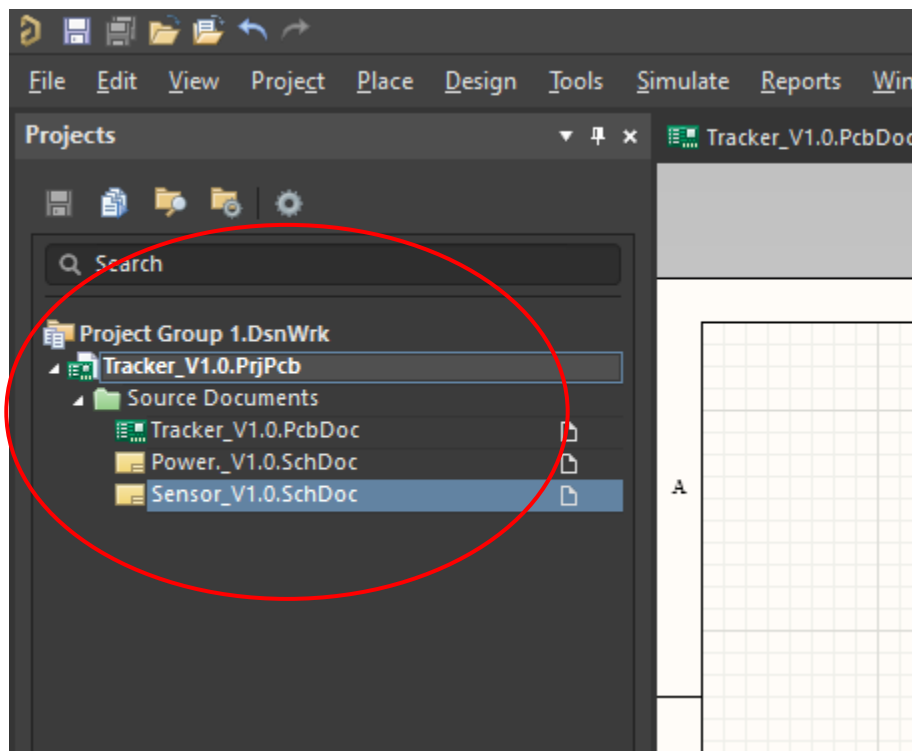
3.6 Numbering Schematic Sheets

Select menu *Tool* -> *Annotation* and *Number Schematic Sheets*, click on *Auto Sheet Number*, *Auto Document Number Update* and *Update Sheet Count*.

3.7 Saving the Project

In the main menu file choose **Save Project As**, enter project name and choose the right destination folder.

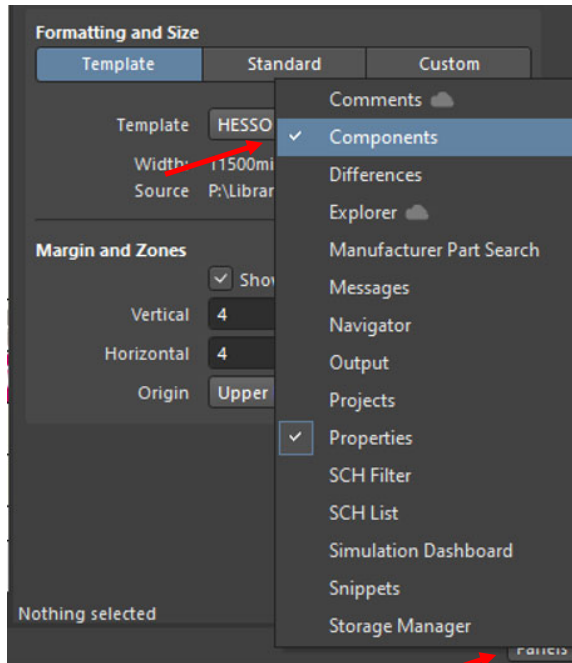
Altium will ask you a name for the *****.SchDoc**. Give an explicit name to the different sheets. Now your Project should look like:



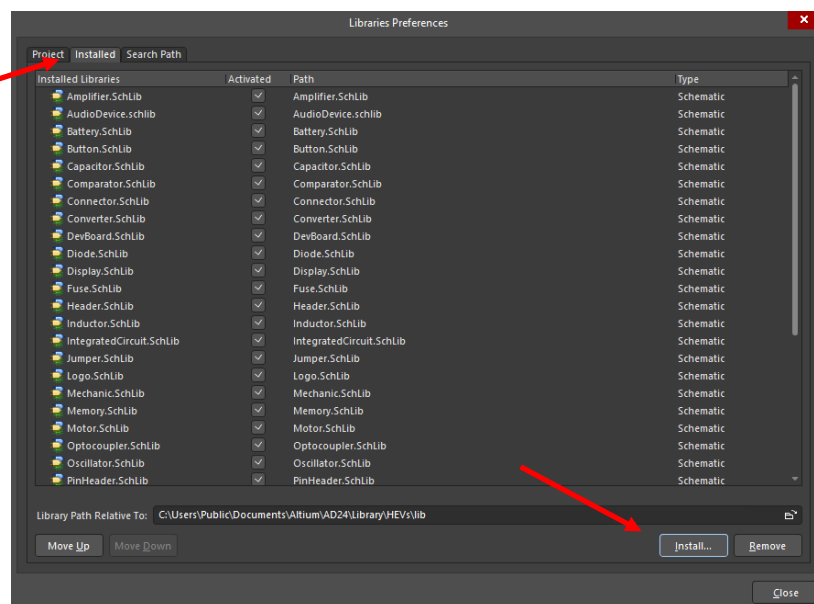
4. Schematic capture

4.1 Setting the Libraries

Click *Panels->Components* on the bottom right of your screen and click the Icon  and choose the pan “Libraries Preferences”.



Choose path : **C:\Users\Public\Documents\Altium\AD24\Library\HEVs\lib** and select all the files. Don't use the other libraries.



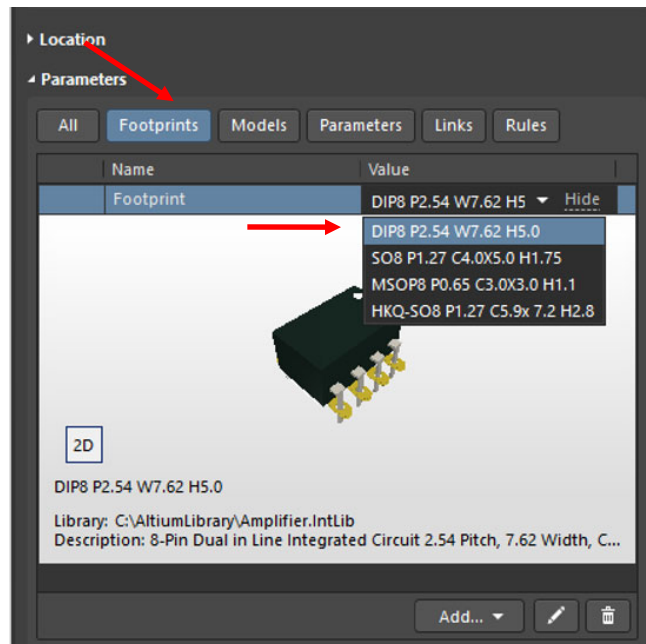
4.2 Placing Parts

Now you can start placing your components, browse the libraries, chose the component you want and drag and drop it to your schematic. Alternately you can use the shortcut key **P-P** (Place-Part...).

Use the **Spacebar** to rotate components, and **X, Y** keys to flip them.

You can edit the component by double clicking on it and you can choose the right pattern.

Don't forget to add mechanical parts as screws for example.



4.3 Wiring Components

Use button  to draw connections between components.

Alternately you can use the shortcut **P-W** (Place-Wire).

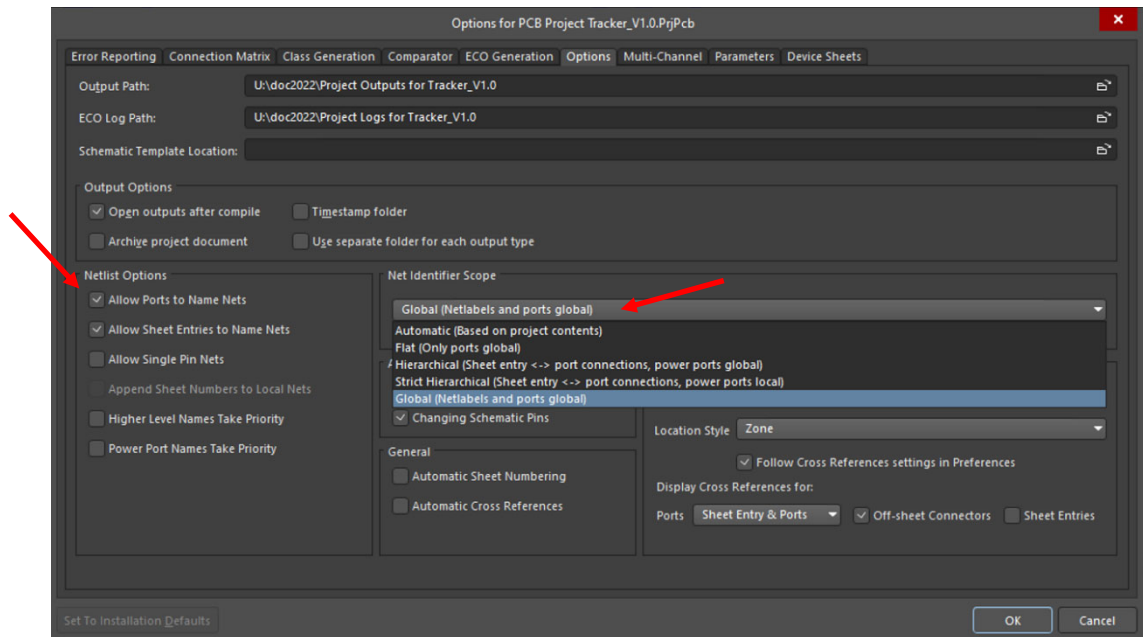
It is useful to keep a large grid. The grid can be changed with the **G** key.

4.4 Ports and Labels

There is different connection method in *Project->Project Options->Options->Net Identifier Scope*.

Use it carefully, in some method nets are not connected between the schematic sheets.

It's preferable to check the option **Allow Ports to Name Nets**, it is useful during the pcb design.



Use button  to place Port, the corresponding shortcut is **P-R** (Place-Port).

Use buttons  to place GND and VCC Power Port.
The corresponding shortcut is **P-O** (Place-Power Port).
You can edit the shape and the name of ports by double clicking on it.
Use the shortcut **P-N** (Place-Net Label) for identify your nets names.

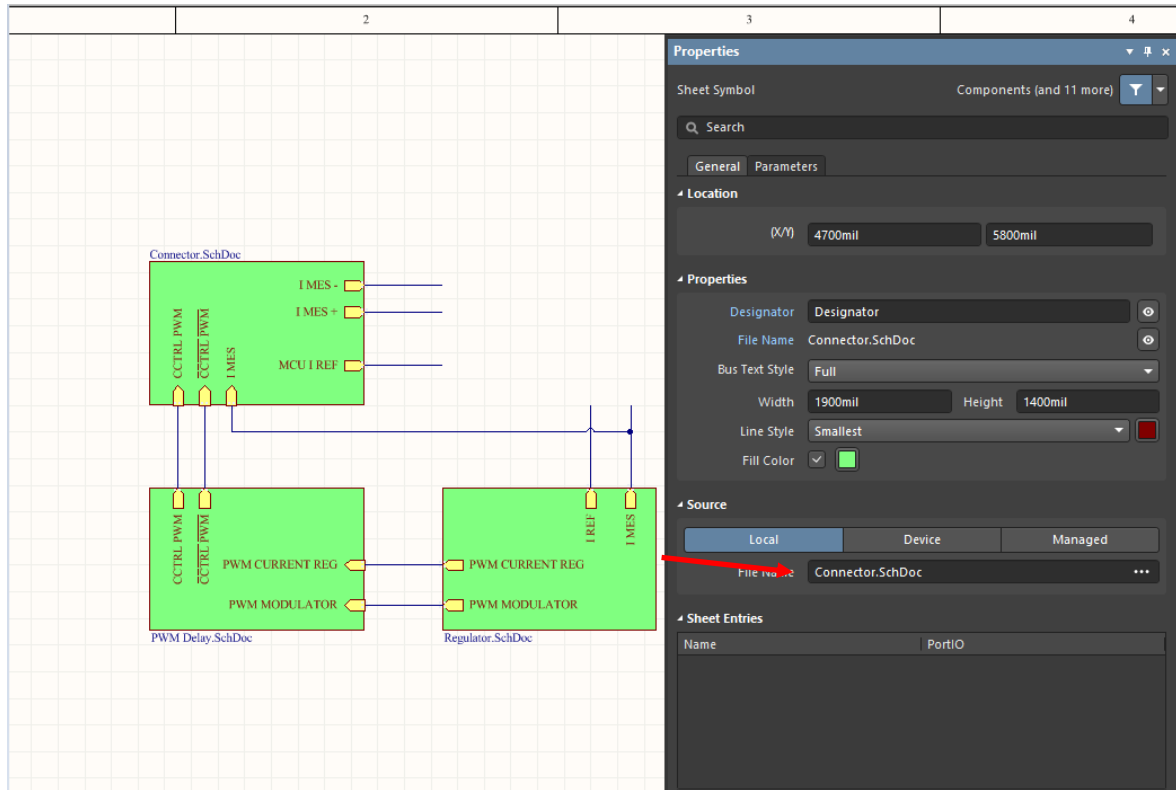
4.5 Schematic Annotation

Use the command *Tools->Annotation->Annotate Schematics Quietly* if you don't need any special features.

With the command *Tools->Annotation->Annotate Schematics...* you can choose between some different annotation methods.

4.6 Hierarchical Schematic

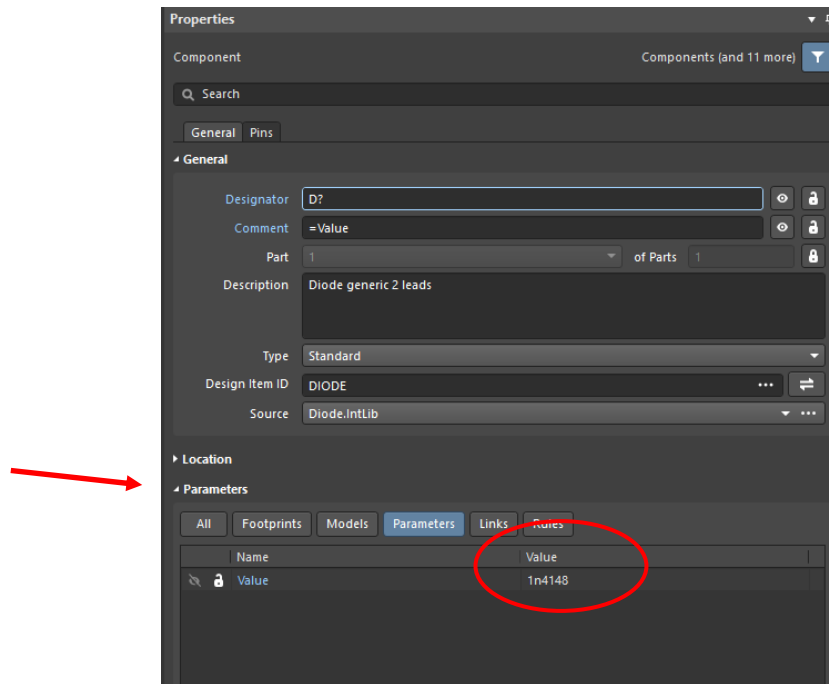
For placing an Entire Schematic Sheet use the command *Place-> Sheet Symbol* or the button . Double click on the green square and choose the linked schematic sheet you want to link.



Use the command *Place-> Sheet Entry* to make a connection between the different sheets.

4.7 Enter the Parts Values

Double Click on a part and edit the tab Parameter to enter the value



Note that the values 100n, 0.1u, 100nF, 0.1uF will be interpreted as different values generating several entries in the bill of material. It is therefore important to always use exactly the same value string for the same components. Moreover 100n can be interpreted as 100nF or 100nH.

A good practice is to use a clear and full numbering with units (except for resistor) as shown below:

- 100nF
- 1uF
- 4.7uF
- 10mF
- 100mH
- 1.3k
- 0.1R
- 1.2R

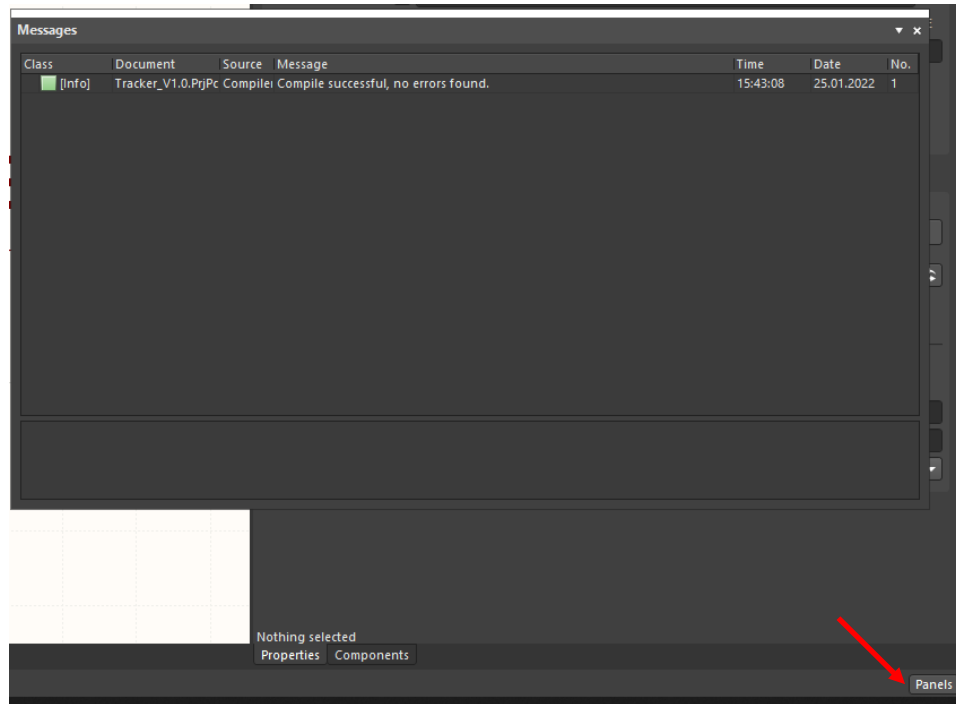
4.8 Schematic Validation

To validate the Schematic right click on your *****.PrjPcb** file and select **Validate PCB Project**

*****.PrjPcb**

If there is error during the compilation a message windows will appear.

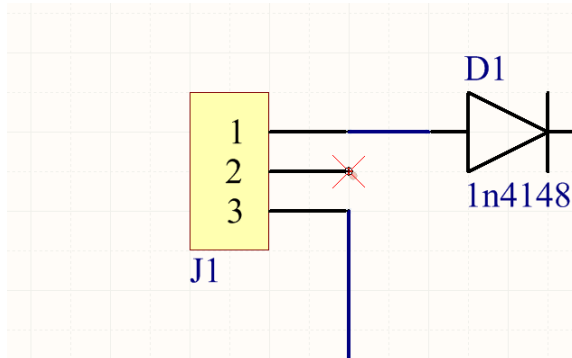
This window doesn't open automatically in case of warning. However, it is advisable to have a look at this message windows by open the **Panels> Messages**



4.9 Function NoERC

It's possible to hide unwanted compilation warning/error by placing a NoERC mark on components pins.

To do this, use the menu *Place->Directives->Generic No Erc*



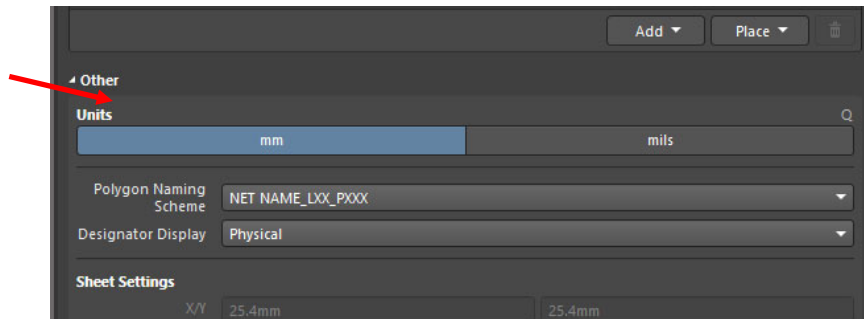
4.10 Print

To print the schematic, the easiest way is to use the menu *File->SmartPDF*.

5. PCB Design

5.1 Grid Setting

Choose between Metric and Imperial by double clicking somewhere on the board.



Use the G key to modify the grid settings. Prefer a large grid, for example 1mm.

5.2 Defining Origin

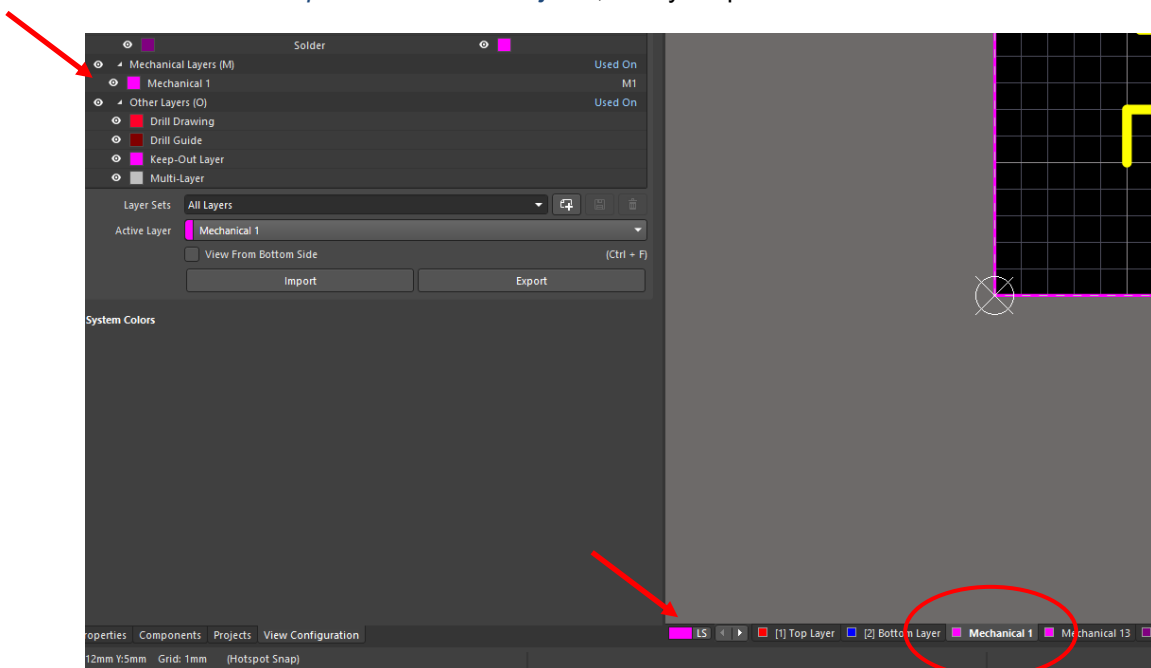
You can change the origin of the grid by selecting *Edit->Origin->Set*. Click the location where you want the 0.0 point, commonly the bottom left of your pcb.

5.3 Drawing Board Outline

Set an adapted grid with the **G** key. Double click on the LS button (Layer Stack) and ensure that Layer Mechanical 1 is visible.

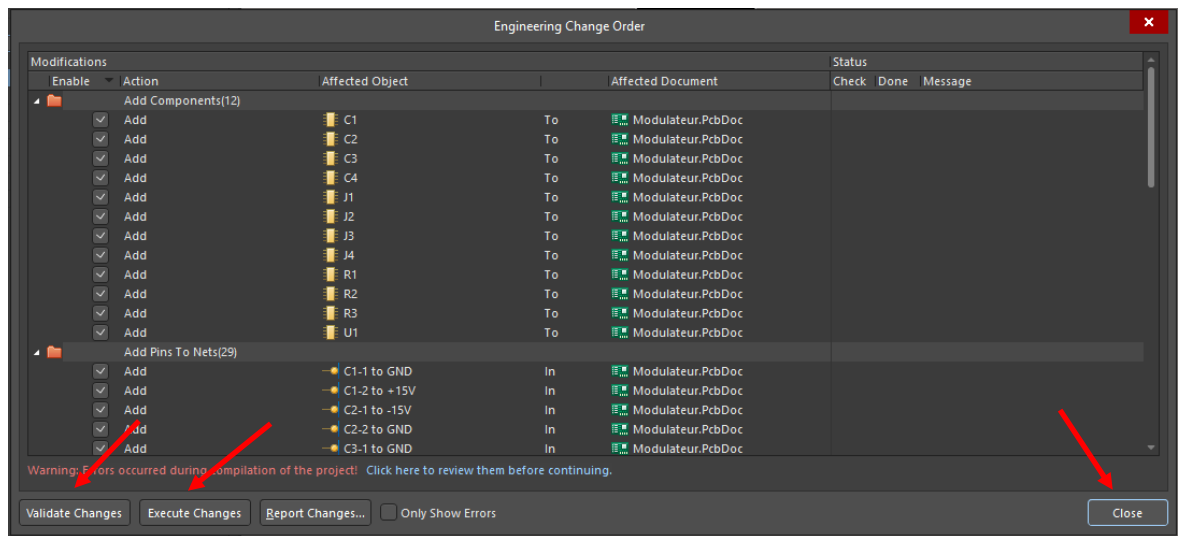
Draw a new board outline on Mechanical 1 layer using the command Place-Line (**P-L**). To edit the line parameter, press the **TAB** key.

Select all the lines on Layer Mechanical 1 and go to the menu *Design->Board Shape->Define Board shape from selected objects*, now your pcb have the new dimensions.



5.4 Importing the Components

Select **Design->Import Change from ***.PrjPcb**, click **Validate Changes**, **Execute Change**, then **Close**.

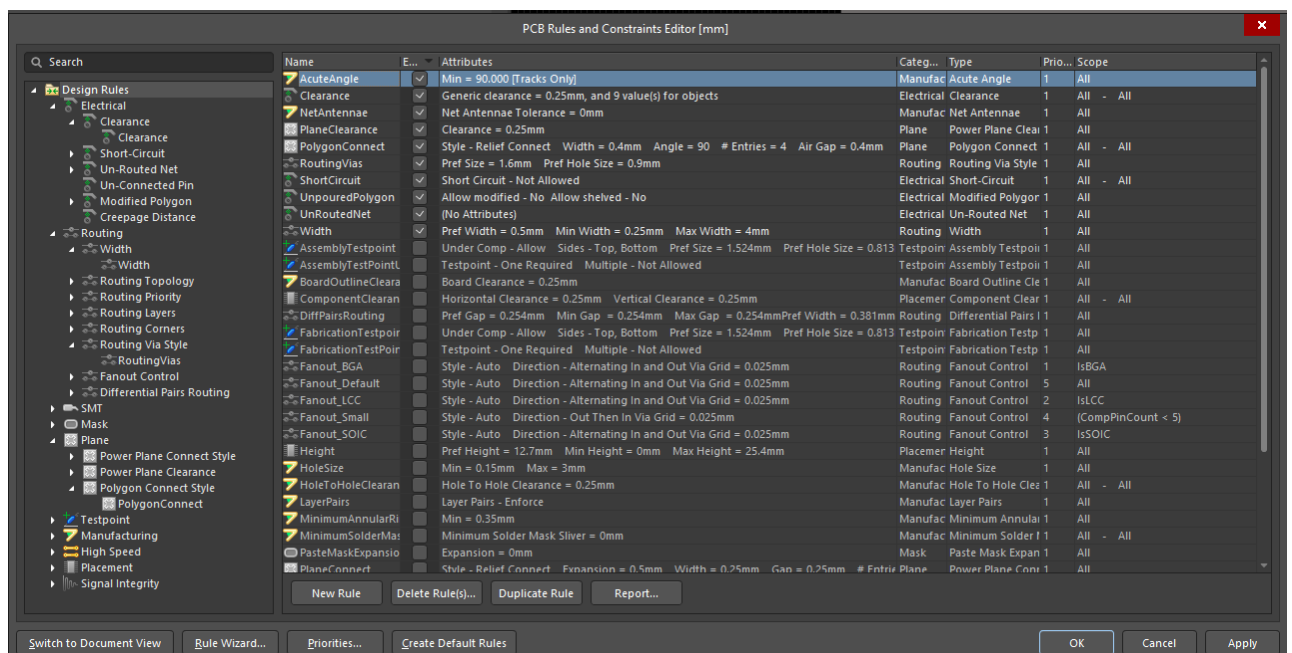


5.5 Move and Place Components

The components can be moved using the mouse. Use the spacebar to rotate components, and the **L key** to change the component Layer. The **G key** allows changing the grid setting: a large grid is preferable for placing components.

5.6 Setting the Design Rules

Open Menu **Design->Rules**, then enable and set the base rules as show in picture below:



5.7 Placing Tracks

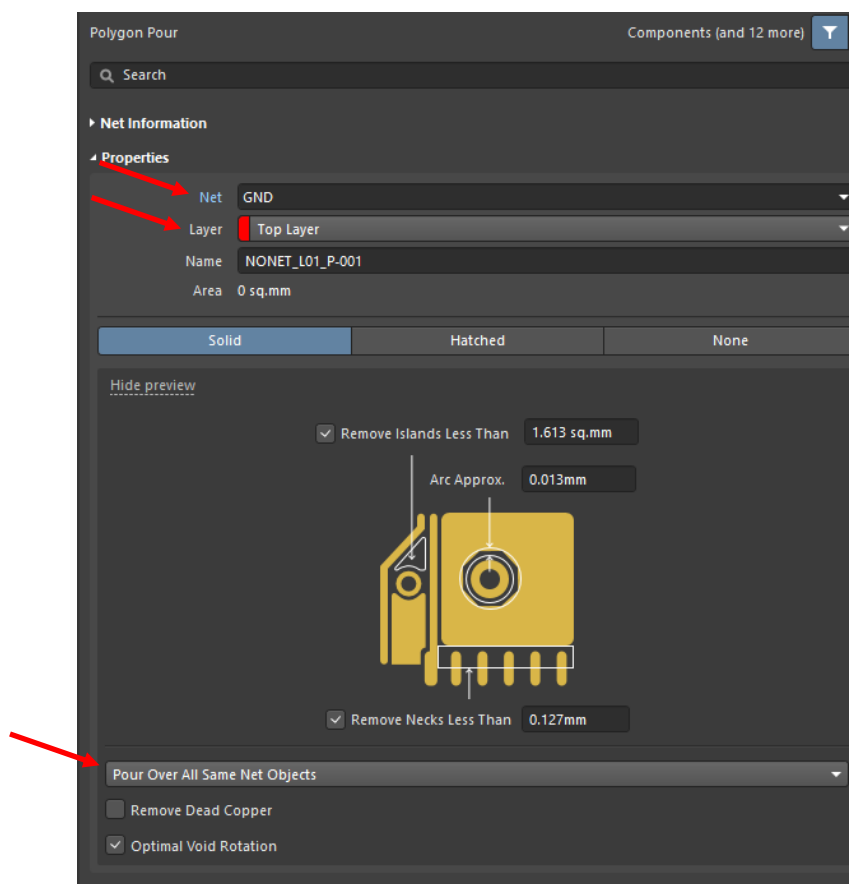
Use the Button  or short-cut **P-T** (PlaceTrack) to draw the tracks.
Push **Tab Key** if you want to edit routing parameter.
Push *** key** on the numeric pad to change routing layer.
Push **shift + Spacebar** for change routing style.
Push **Space-bar** for change the angle orientation.

5.8 Placing Vias

To add a via simply push the *** key** on the numeric pad during routing,
or the shortcut **P-V** (Place Via).
HESS-SO via capability is v160h90 (Pad 1.6mm Hole 0.9mm).

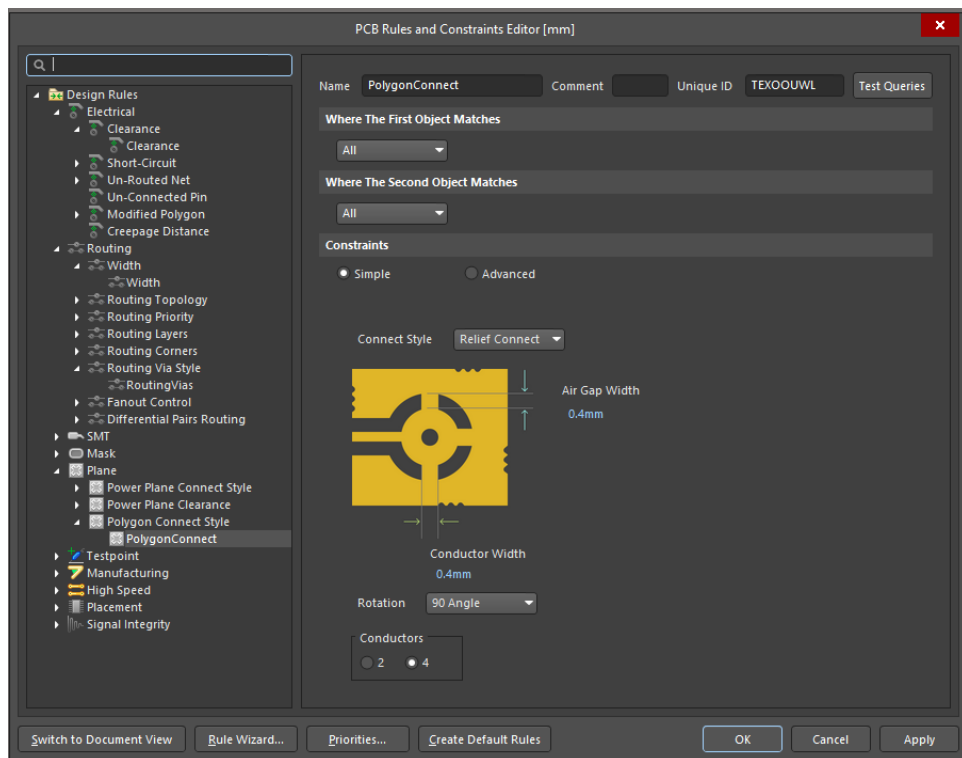
5.9 Placing Polygon Pour

Use the menu [Place->Polygon Pour](#) or the shortcut **P-G** (Place Polygon Pour) to draw a copper pour.
Select the layer, the net you want to pour and the connections style.

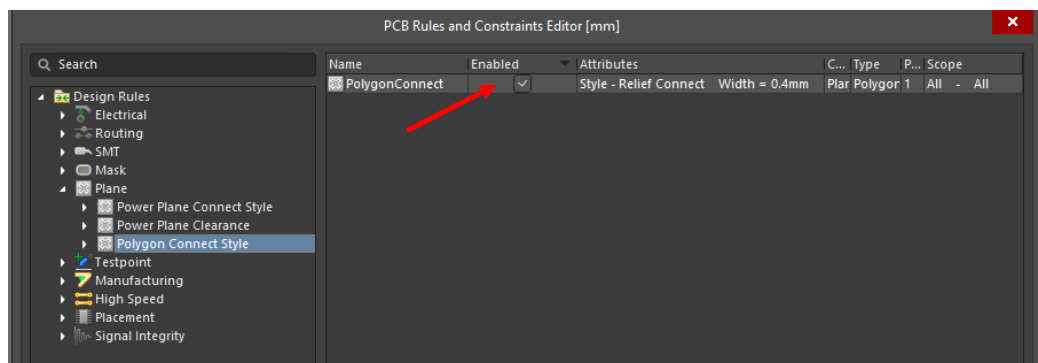


Altium Designer

Select the connections style with *Design->Rules->PolygonConnect*



Ensure that the “Polygon Connect” rule is enabled.

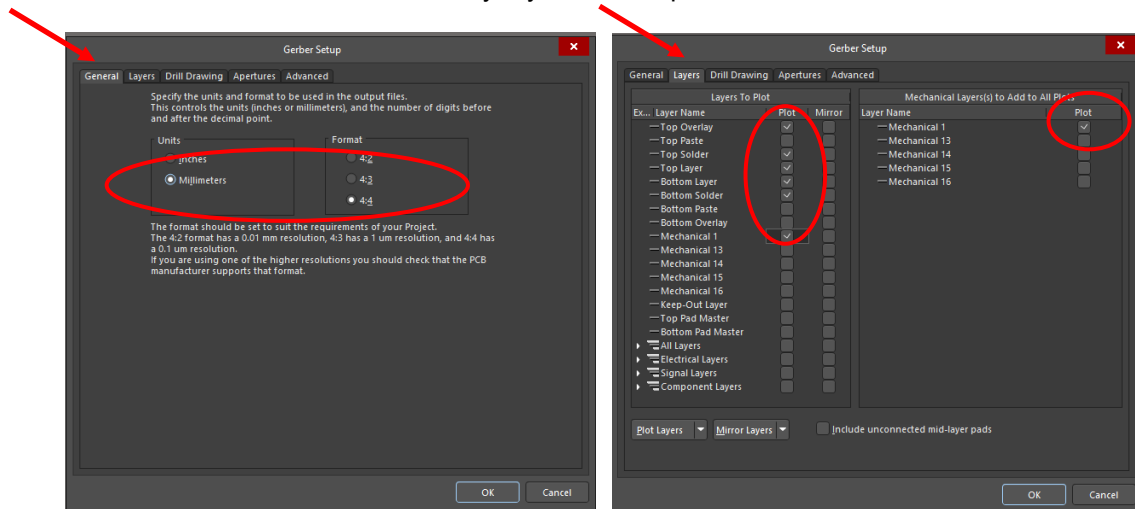


5.10 Design Rules Check

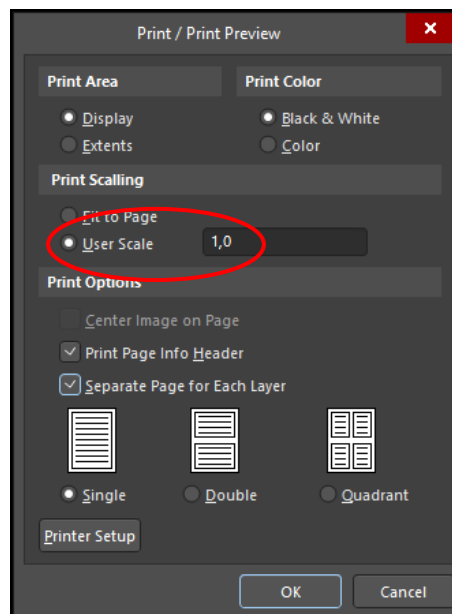
Check your PCB with function *Tools->Design Rule Check-> Run Design Rules Check*.
Check the report and correct de actual errors.

5.11 Files Output

Use the menu *File->Fabrication Outputs-> Gerber Files*.
Select Millimeters 4:4, and which layer you want to print



Altium open a new CAMtastic window, select *File->Print* if you want to print your layer.
Scale must to be set to 1.0.



5.12 Print Drill Table

Select the Menu *Place->Drill Table*, and then the Menu *File->Fabrication Output->Drill Drawing* and use the Print button.

6. Tips & Tricks

6.1 Help

For open help during an action hit the **F1** key.

6.2 3D

Push key **3** or key **2** for switch between 3D or 2D view.

6.3 Zoom

You can easily zoom IN & OUT with **CTRL** Key + Right Mouse + Moving Mouse UP Down.

6.4 Selecting multiple objects

Right click on an Object, choose **Find similar Objects**, edit all object with PCB Inspector (F11).

6.5 Setting properties

Pushing the **TAB** key during an action allow to edit some properties